

Project 1 - Sprint 2 - Description

1. Sprint 2

The goal in this second sprint is creating a network simulation for the structured cabling project developed in the first sprint. In this sprint the focus is on the layer two infrastructure, and the layer three fundamentals (IPv4 addressing, and static routing).

For such purposes, the teams should use the **Cisco Packet Tracer** network simulation tool.

1.1. Teamwork organization

Each team member will keep working over the same terminal, the task is now creating a Packet Tracer simulation of the networks on the floors covered by the structured cabling project developed in the first sprint.

Each of these simulations will encompass a single terminal, however, **they must include the campus backbone**. As we will see ahead, this means all simulations must include a switch representing the main cross-connect, and switches representing each of the other terminals (intermediate cross-connects).

The team member assigned with **Terminal 3** has one additional final task: putting together into a single Packet Tracer simulation all the simulations created by each team member. If there is a good coordination between team members work, this is just a simple copy and paste operation between Packet Tracer instances. The best option for cutting points is the connection of each terminal to the campus backbone network, each terminal's main switch should be kept and the backbone connections between those switches rebuilt.

1.2. Packet Tracer version

To avoid any issues when putting together all team members' work into a single Packet Tracer simulation, all team members should use the same Packet Tracer version. They all should use the version suggested in the laboratory classes. This should be established in the planning meeting, and registered in the **planning.md** document.

1.3. Devices naming

When a new device is added to the Packet Tracer simulation, a display name is automatically assigned, but then it should be changed to a meaningful name. During the planning meeting, the team should discuss some rules regarding this, it will facilitate the teamwork, and this should be included in the **planning.md** document.

The display name is just for Packet Tracer to display, at the OS level, each device has a hostname (DNS name), that can be settled by using the **hostname** command. Mind in DNS names, no spaces are allowed.

2. Layer two configuration

The goal is creating a logical and not a physical network layout, the layout should nevertheless be representative of the structured cabling project from the preceding sprint.

Specifically:

- Every cross-connect (distributor), including consolidation points (CP), in the structured cabling project is to be represented in the simulation by a layer two network switch. You may use the **PT-empty** switch model and add the modules with the required port types.
- Cable types interconnecting different cross-connects (switches in the simulation) must match the cable types (copper or fibre) defined in the structured cabling project, and the same goes for horizontal cabling subsystems.
- Path redundancy cable links between cross-connects of the same level (e.g., between two intermediate cross-connects), present in the structured cabling project should as well be present in the simulation. Nevertheless, multiple cable links (for link aggregation) between two cross-connects of distinct levels (e.g., between an intermediate cross-connect and the main cross-connect) should be represented as a single cable link between two switches in the simulation.

2.1. Virtual LANs

Several VLANs are required for traffic separation, some of them are for specific use in some terminals, and others are meant to be shared between terminals. Nevertheless, **all VLANs must be available on every switch at every terminal**, even if they are not locally used because they are for specific use in another terminal.

2.1.1. VLANs used by all terminals

- Campus backbone – used to transfer traffic between terminals, each terminal has a router with an IP address on this network.
- Switches DMZ – used for administration of layer two switches, each and every switch has an IP address on this network.

2.1.2. VLANs used by specific terminals

In each terminal, the following four VLANs are required:

- User outlets – work area outlets.
- Wi-Fi – Wi-Fi network (for all access-points' outlets within the terminal).
- VoIP – VoIP outlets (for all IP-phones within the terminal).
- Servers DMZ – Network servers within the terminal.

Despite being associated to a specific terminal, **these VLANs must be available on every switch in every terminal**, meaning all and every switch should have all terminals' VLANs in the local VLAN database and all switches' interconnections must be configured in trunk-mode with all the VLANs.

So, **to summarize and clarify**, for a project including the four terminals (teams with four members), there will be a total of:

$$2 + (4 \times 4) = 18 \text{ VLANs}$$

If the team decides not to use any of those as the default VLAN, there will be also the **default VLAN**, teams may keep the use of VLANID 1 for the default VLAN.

2.2. Spanning tree protocol

If your layout encompasses redundant cable links between switches, then STP is essential. Switches have STP enabled by default, so there is nothing to do here, simply **do not disable STP on any switch**.

2.3. VLAN Trunking Protocol (VTP)

This protocol is one very convenient way to ensure all switches have the same complete VLAN database. To achieve that:

- All switches must use the same VTP domain name (**vtp domain ...**).
- At least one switch must be in server mode (**vtp mode server**). These are the switches where the VLAN database may be manually defined (**vlan**) and changed whenever needed.

As far as switches interconnection ports are in trunk mode (**switchport mode trunk**), and the VTP domain name is the same, VTP will ensure the VLAN database established by the servers is propagated to every switch of the domain operating in either VTP server mode or VTP client mode (**vtp mode client**).

2.4. The VLAN database

Within the team, all Packet Tracer simulations must use the same VLAN database (manually established on VTP server switches). As result from the planning meeting, there should be a table or list of VLANs to be used by all team members, this must be included in the **planning.md** document.

Each VLAN should have a unique meaningful name, and of course a unique VLANID. In Cisco devices VLANIDs between 2 and 1000 are free to be used without interfering with other features. **For this project, each team is required to use a specific range of values for VLANIDs**, please check the last chapter, **“Per team specific data - mandatory use by teams”**.

2.5. VTP domain

VTP only propagates the VLAN database between switches belonging to the same VTP domain, thus, so that later all terminals' simulations may be putted together easily, all team members must use the same VTP domain name on every switch. The VTP domain name can be 1 to 32 characters long, spaces are not supported. The VTP domain name for each team is established at the **“Per team specific data - mandatory use by teams”** chapter in the last pages of this document. The adopted VTP domain name should also be registered in the **planning.md** document.

3. Layer three configuration

Over the established layer two infrastructure, IPv4 networks will then be defined, usually, **one IPv4 network for each VLAN**.

The IPv4 traffic forwarding between VLANs is implemented by **a router in each terminal**. In each terminal, the corresponding router is connected to the Campus Backbone VLAN and provides access to the networks within the terminal.

3.1. IPv4 networks

The IPv4 network address assigned to each VLAN must meet the following requirements regarding the maximum number of IPv4 nodes it must be able to support.

3.1.1. Campus backbone and Switches DMZ

- Campus backbone: 190 nodes
- Switches DMZ: 410 nodes

3.1.2. Terminal 2

- User outlets: 1000 nodes.
- Wi-Fi: 2000 nodes.
- VoIP: 500 nodes.
- Servers DMZ: 100 nodes.

3.1.3. Terminal 3

- User outlets: 950 nodes.
- Wi-Fi: 1500 nodes.
- VoIP: 350 nodes.
- Servers DMZ: 150 nodes.

3.1.4. Terminal 5

- User outlets: 1900 nodes.
- Wi-Fi: 4000 nodes.
- VoIP: 500 nodes.
- Servers DMZ: 200 nodes.

3.1.5. Terminal 4

- User outlets: 800 nodes.
- Wi-Fi: 1800 nodes.
- VoIP: 400 nodes.
- Servers DMZ: 120 nodes.

Each team has a specific **block of IPv4 addresses that must be used**, check the last chapter of this document: **“Per team specific data - mandatory use by teams”**.

For each team, all used IPv4 network addresses must be non-overlapping, and they are required to belong to the provided block of addresses.

In the sprint planning, the team will establish:

- The IPv4 network address assigned to the Campus Backbone network.
- The IPv4 node address each router will use for its Campus Backbone network connection.
- The IPv4 network address assigned to the Switches DMZ network.
- The IPv4 node addresses range, within the Switches DMZ network, each terminal will use for its switches.
- The block of addresses each team member will use for the networks specific to its terminal. During the sprint, the team member in charge of each terminal will use this block of addresses to assign IPv4 addresses to the networks within the terminal.

Important: addresses blocks assigned to each team member can't overlap each other's, and they cannot overlap either with the IPv4 network address assigned to the Campus Backbone network or the Switches DMZ network.

The team master is responsible for registering this planning information and make it available to the team (**planning.md**).

3.2. End devices in the Packet Tracer simulation

For each VLAN **in use** at a terminal, there must be at least one end-device connected to that VLAN. So, in the Packet Tracer simulation, for each terminal there should be at least:

- A workstation (PC) connected to the User Outlets VLAN IPv4 network of the terminal.
- A workstation (PC) connected to the Switches DMZ IPv4 network.
- A Wireless laptop, associated with a wireless access-point (not a wireless router) connected to the Wi-Fi VLAN IPv4 network of the terminal.
- A server connected to the Servers DMZ VLAN IPv4 network of the terminal.
- A VoIP phone connected to the VoIP VLAN IPv4 network of the terminal.

End devices must be connected to the correct VLANs by configuring the corresponding switch ports in access mode (**switchport mode access**) and assigning to them the correct VLANs (**switchport access vlan ...**).

Except for the VoIP phones, all other end devices are required to have a static and manually defined IPv4 configuration, including the default gateway.

For VoIP phones, the **7960 model** is to be used, the corresponding port on the switch must also be configured in access mode (**switchport mode access**), however, the access VLAN must be disabled (**no switchport access vlan**) and the voice VLAN must be used instead (**switchport voice vlan ...**).

The IPv4 configuration of the VoIP phones is out of scope for now and is addressed on the next sprint.

3.3. Switches remote management

Switches operate by retransmitting layer two frames and are not required to have an IP address, nevertheless, most switches have remote management capabilities, and those require the switch to have at least one IP address.

By using specific application protocols like for instance SSH, SNMP and HTTP, administrators and monitoring systems can use the network to access and manage switches and other network devices.

One very recommended practice for remote management of network devices is having an independent and isolated network for such purpose. **In this project the Switches DMZ is a dedicated and isolated network (without routers/gateways) in which every switch has one IPv4 address.**

Unlike what happens in routers and other layer three devices, in switches IP addresses are not assigned to physical interfaces, instead they are assigned to VLAN interfaces. As far as a VLAN is available at a switch, it is possible to use a virtual network interface connected to it (**interface vlan ...**) and establish the corresponding IP configuration (**ip address ...**).

For instance, if in your case the Switches DMZ is using VLANID **500**, and you want to assign to the switch the IPv4 node address **172.32.56.43/23**, the sequence of configuration commands is:

```
interface vlan 500
ip address 172.32.56.43 255.255.254.0
no shutdown
```

The Switches DMZ is meant to be an isolated network, meaning traffic from other networks is not allowed to enter it and traffic inside this network is never transferred to other networks. To access and manage the switches, each terminal has a dedicated workstation connected to this network.

3.4. Routers and static routing

In each terminal there must be a router. **Please use a 2811 model, and no other model**, other router models may not support VoIP in Packet Tracer, and that is required in the next sprint.

The router in each terminal will assure IPv4 traffic forwarding between local IPv4 networks (within the terminal, except for the Switches DMZ) and other terminals' IPv4 networks. Traffic to other terminals is routed through the Campus Backbone IPv4 network, to the appropriate router at the target terminal.

IPv4 node addresses being used by every router on the backbone network are established in the planning meeting. Together with the blocks of addresses assigned to each terminal, they make it possible to build the static routing table on every router.

Each team member is assigned the task of creating the Packet Tracer simulation for one terminal. Each of these simulations will encompass a single terminal, however, **they must include the campus backbone**. The main cross-connect and all intermediate cross-connects must be represented by a switch, and a router (2811 model) must be connected to each.

Because each router is connected to five different IPv4 networks, one might expect a router with five network interfaces would be required. However, the same way a single trunk-mode connection between two switches can connect several different VLANs, for a connection to a router the same applies, and on the router side each VLAN appears as different logical network interface (sub interface).

So, to connect the router to those five networks, a single cable connection to a trunk-mode port in the switch is the best solution.

Regarding the configuration of routers outside the terminal, it should encompass only the connection to the backbone VLAN and corresponding IP address. The full configuration of each terminal's router is up to the team member in charge of that terminal.

3.5. Internet connection

Terminal 2 has an internet connection. The member in charge of this terminal (T.2.1.) will represent it by a DSL modem connection to an ISP router. The IPv4 node address in use by the ISP router is different for each team, it's available at the **"Per team specific data - mandatory use by teams"** chapter in the last pages of this document.

The ISP router must have a static routing table such as it will forward the matching traffic into the infrastructure.

4. Sprint 2 backlog

Task	Task description
T.2.1	Development of a layer two and layer three Packet Tracer simulation for Terminal 2 encompassed floors, the campus backbone, and including the internet connection.
T.2.2	Development of a layer two and layer three Packet Tracer simulation for Terminal 3 encompassed floors and the campus backbone. Integration of every member's Packet Tracer simulation into a single simulation.
T.2.3	Development of a layer two and layer three Packet Tracer simulation for Terminal 5 encompassed floors and the campus backbone.
T.2.4	Development of a layer two and layer three Packet Tracer simulation for Terminal 4 encompassed floors and the campus backbone.

Task T.2.4 is not applicable to teams with only three members.

5. Sprint 2 outputs/products

For each task on this sprint, the main output is the Packet Tracer simulation file for the corresponding terminal, it should be named **TerminalN.pkt**, with N replaced by the number identifying the terminal. Each team member is to commit that file into the personal sprint folder at the repository.

A document (in any standard format) detailing how IPv4 network addresses were established and how static routing tables were created is also required.

Each team member must also commit to the personal sprint folder a text file with a configuration dump for every switch and every router within the encompassed terminal. These configuration text files can be easily exported in Packet Tracer: within the device's window click the **export** button on the **Running Config**, or the **Startup Config** (if the **Running Config** has been saved).

The default name for the text file may be kept, as it represents the device's display name. During the sprint, members are to commit changes to these files as change are made to their devices' configurations, this allows the teacher to follow the progresses along the sprint. These files are also a safeguard, if somehow the configuration of a device in Packet Tracer is lost, it may be restored from these files.

For task **T.2.2** there is one additional output to be committed into the personal sprint folder, it is the overall simulation integrating all members work, that file should be named **campus.pkt**. Regarding configuration files, for task T.2.2, they only encompass the devices in Terminal 3.

6. Sprint 2 planning

- One member takes the role of **sprint master** for sprint 2.

On each sprint, a different team member takes this role.

Beyond a technical consultant, the class teacher also assumes the project owner role. Any doubts regarding the provided sprint description and the project in general should be addressed to the class teacher.

The sprint master will insert a summary of the backlog in file **/doc/sprint2/planning.md**.

- Global technical decisions and team coordination.

Even though the tasks established in the sprint backlog are rather independent, as they all belong to the same project, a coordination effort in this phase is key. This will most often encompass technical decisions regarding the implementation with a special focus on features that are shared between tasks.

All taken technical decisions must be registered, by the sprint master, in file **/doc/sprint2/planning.md**.

For this specific sprint, some decisions that must be settled as soon as possible are:

- The Packet Tracer version to be used on this sprint (and following sprint). All team members should use the same version. Each team member is assigned a task, and for each task one independent simulation will be created, but later they must all be putted together in a single simulation and that could raise some issues if different Packet Tracer versions were used.
 - VLANIDs and corresponding VLAN names to be used, they must be unique. Don't forget to establish also the native VLAN (untagged) and the default VLAN, probably teams will want to keep the default in Cisco devices (VLANID=1). **For each team there is a range of VLANID values that can be used.**
 - The VTP domain name to be used. When all simulations are putted together, all switches belonging to every terminal will keep the same VLAN database by talking to each other's, so the VTP domain must be the same on every switch. **For each team there is an already assigned VTP domain name.**
 - The IPv4 network address for the Campus Backbone network, this network address must belong to the address space provided to the team (addresses block), and yet it cannot overlap any used network or blocks of addresses assigned to team members for use within each terminal.
 - The IPv4 network address for the Switches DMZ network, this network address must belong to the address space provided to the team (addresses block), and yet it cannot overlap any used network or blocks of addresses assigned to team members for use within each terminal.
 - The IPv4 block of addresses assigned to each team member. All these blocks must belong to the address space provided to the team. They must be carefully set by the team so that they are valid, non-overlapping, and they provide the required number of addresses for each terminal.
 - A unique IPv4 node address in the Campus Backbone network for each team member. This IPv4 node address is intended to be used by the terminal's router connection to the backbone.
 - A unique range of IPv4 node addresses within the Switches DMZ network for each team member. This range of IPv4 addresses is intended to be used by the terminal's switches.
- Tasks assignment to team members.

Every member (sprint master included) is assigned exactly one task from the backlog. The provided backlog has seven tasks, for teams with less than 4 members, the last tasks in the backlog are to be ignored, the first ones are mandatory.

The sprint master registers in file **/doc/sprint2/planning.md** the task assigned to each team member.

- For this sprint, the tasks are the follow out of previous sprint's tasks, thus they are already assigned. This means, a team member that was assigned in the previous sprint the structured cabling project of some terminal is now assigned the task of creating the Packet Tracer layer two and layer three simulation for that same terminal.
- Tasks that were not assessed as ***Totally implemented with no issues***, should be completed, and have issues fixed by the same team member during this sprint. This must also be registered in the **planning.md** document by the sprint master.

7. Per team specific data - mandatory use by teams

7.1. VLANID and VTP domain names each team must use

Class	Team number (within class)	VTP domain name to use	VLANIDs range to use
2DA	1	rc2526dag1	740 to 760
	2	rc2526dag2	741 to 761
	3	rc2526dag3	742 to 762
	4	rc2526dag4	743 to 763
	5	rc2526dag5	744 to 764
	6	rc2526dag6	745 to 765
	7	rc2526dag7	746 to 766
2DB	1	rc2526dbg1	747 to 767
	2	rc2526dbg2	748 to 768
	3	rc2526dbg3	749 to 769
	4	rc2526dbg4	750 to 770
	5	rc2526dbg5	751 to 771
	6	rc2526dbg6	752 to 772
	7	rc2526dbg7	753 to 773
2DC	1	rc2526dcg1	754 to 774
	2	rc2526dcg2	755 to 775
	3	rc2526dcg3	756 to 776
	4	rc2526dcg4	757 to 777
	5	rc2526dcg5	758 to 778
	6	rc2526dcg6	759 to 779
	7	rc2526ddg7	760 to 780
2DD	1	rc2526ddg1	761 to 781
	2	rc2526ddg2	762 to 782
	3	rc2526ddg3	763 to 783
	4	rc2526ddg4	764 to 784
	5	rc2526ddg5	765 to 785
	6	rc2526ddg6	766 to 786
	7	rc2526ddg7	767 to 787
2DE	1	rc2526deg1	768 to 788
	2	rc2526deg2	769 to 789
	3	rc2526deg3	770 to 790
	4	rc2526deg4	771 to 791
	5	rc2526deg5	772 to 792
	6	rc2526deg6	773 to 793
	7	rc2526deg7	774 to 794
2DF	1	rc2526dfg1	775 to 795
	2	rc2526dfg2	776 to 796
	3	rc2526dfg3	777 to 797
	4	rc2526dfg4	778 to 798
	5	rc2526dfg5	779 to 799
	6	rc2526dfg6	780 to 800
	7	rc2526dfg7	781 to 801
2DG	1	rc2526dgg1	782 to 802
	2	rc2526dgg2	783 to 803
	3	rc2526dgg3	784 to 804
	4	rc2526dgg4	785 to 805
	5	rc2526dgg5	786 to 806
	6	rc2526dgg6	787 to 807
	7	rc2526dgg7	788 to 808

Class	Team number (within class)	VTP domain name to use	VLANIDs range to use
2DH	1	rc2526dhg1	789 to 809
	2	rc2526dhg2	790 to 810
	3	rc2526dhg3	791 to 811
	4	rc2526dhg4	792 to 812
	5	rc2526dhg5	793 to 813
	6	rc2526dhg6	794 to 814
	7	rc2526dhg7	795 to 815
2DI	1	rc2526dig1	796 to 816
	2	rc2526dig2	797 to 817
	3	rc2526dig3	798 to 818
	4	rc2526dig4	799 to 819
	5	rc2526dig5	800 to 820
	6	rc2526dig6	801 to 821
	7	rc2526dig7	802 to 822
2DJ	1	rc2526djg1	803 to 823
	2	rc2526djg2	804 to 824
	3	rc2526djg3	805 to 825
	4	rc2526djg4	806 to 826
	5	rc2526djg5	807 to 827
	6	rc2526djg6	808 to 828
	7	rc2526djg7	809 to 829
2DK	1	rc2526dkg1	810 to 830
	2	rc2526dkg2	811 to 831
	3	rc2526dkg3	812 to 832
	4	rc2526dkg4	813 to 833
	5	rc2526dkg5	814 to 834
	6	rc2526dkg6	815 to 835
	7	rc2526dkg7	816 to 836
2DL-E	1	rc2526dlg1	817 to 837
	2	rc2526dlg2	818 to 838
	3	rc2526dlg3	819 to 839
	4	rc2526dlg4	820 to 840
	5	rc2526dlg5	821 to 841
	6	rc2526dlg6	822 to 842
	7	rc2526dlg7	823 to 843
2NA	1	rc2526nag1	824 to 844
	2	rc2526nag2	825 to 845
	3	rc2526nag3	826 to 846
	4	rc2526nag4	827 to 847
	5	rc2526nag5	828 to 848
	6	rc2526nag6	829 to 849
	7	rc2526nag7	830 to 850
2NB	1	rc2526nbg1	831 to 851
	2	rc2526nbg2	832 to 852
	3	rc2526nbg3	833 to 853
	4	rc2526nbg4	834 to 854
	5	rc2526nbg5	835 to 855
	6	rc2526nbg6	836 to 856
	7	rc2526nbg7	837 to 857

7.2. IPv4 addresses to use, and the ISP router's IPv4 address

Class	Team number (within class)	IPv4 address space to use (Block of IPv4 addresses)	ISP router's IPv4 node address
2DA	1	10.47.0.0/17	89.73.67.1/30
	2	10.47.128.0/17	89.73.67.6/30
	3	10.48.0.0/17	89.73.67.9/30
	4	10.48.128.0/17	89.73.67.14/30
	5	10.49.0.0/17	89.73.67.17/30
	6	10.49.128.0/17	89.73.67.22/30
	7	10.50.0.0/17	89.73.67.25/30
2DB	1	10.50.128.0/17	89.73.67.30/30
	2	10.51.0.0/17	89.73.67.33/30
	3	10.51.128.0/17	89.73.67.38/30
	4	10.52.0.0/17	89.73.67.41/30
	5	10.52.128.0/17	89.73.67.46/30
	6	10.53.0.0/17	89.73.67.49/30
	7	10.53.128.0/17	89.73.67.54/30
2DC	1	10.54.0.0/17	89.73.67.57/30
	2	10.54.128.0/17	89.73.67.62/30
	3	10.55.0.0/17	89.73.67.65/30
	4	10.55.128.0/17	89.73.67.70/30
	5	10.56.0.0/17	89.73.67.73/30
	6	10.56.128.0/17	89.73.67.78/30
	7	10.57.0.0/17	89.73.67.81/30
2DD	1	10.57.128.0/17	89.73.67.86/30
	2	10.58.0.0/17	89.73.67.89/30
	3	10.58.128.0/17	89.73.67.94/30
	4	10.59.0.0/17	89.73.67.97/30
	5	10.59.128.0/17	89.73.67.102/30
	6	10.60.0.0/17	89.73.67.105/30
	7	10.60.128.0/17	89.73.67.110/30
2DE	1	10.61.0.0/17	89.73.67.113/30
	2	10.61.128.0/17	89.73.67.118/30
	3	10.62.0.0/17	89.73.67.121/30
	4	10.62.128.0/17	89.73.67.126/30
	5	10.63.0.0/17	89.73.67.129/30
	6	10.63.128.0/17	89.73.67.134/30
	7	10.64.0.0/17	89.73.67.137/30
2DF	1	10.64.128.0/17	89.73.67.142/30
	2	10.65.0.0/17	89.73.67.145/30
	3	10.65.128.0/17	89.73.67.150/30
	4	10.66.0.0/17	89.73.67.153/30
	5	10.66.128.0/17	89.73.67.158/30
	6	10.67.0.0/17	89.73.67.161/30
	7	10.67.128.0/17	89.73.67.166/30
2DG	1	10.68.0.0/17	89.73.67.169/30
	2	10.68.128.0/17	89.73.67.174/30
	3	10.69.0.0/17	89.73.67.177/30
	4	10.69.128.0/17	89.73.67.182/30
	5	10.70.0.0/17	89.73.67.185/30
	6	10.70.128.0/17	89.73.67.190/30
	7	10.71.0.0/17	89.73.67.193/30
2DH	1	10.71.128.0/17	89.73.67.198/30
	2	10.72.0.0/17	89.73.67.201/30
	3	10.72.128.0/17	89.73.67.206/30
	4	10.73.0.0/17	89.73.67.209/30
	5	10.73.128.0/17	89.73.67.214/30
	6	10.74.0.0/17	89.73.67.217/30
	7	10.74.128.0/17	89.73.67.222/30

Class	Team number (within class)	IPv4 address space to use (Block of IPv4 addresses)	ISP router's IPv4 node address
2DI	1	10.75.0.0/17	89.73.67.225/30
	2	10.75.128.0/17	89.73.67.230/30
	3	10.76.0.0/17	89.73.67.233/30
	4	10.76.128.0/17	89.73.67.238/30
	5	10.77.0.0/17	89.73.67.241/30
	6	10.77.128.0/17	89.73.67.246/30
	7	10.78.0.0/17	89.73.67.249/30
2DJ	1	10.78.128.0/17	89.73.67.254/30
	2	10.79.0.0/17	89.77.67.1/30
	3	10.79.128.0/17	89.77.67.6/30
	4	10.80.0.0/17	89.77.67.9/30
	5	10.80.128.0/17	89.77.67.14/30
	6	10.81.0.0/17	89.77.67.17/30
	7	10.81.128.0/17	89.77.67.22/30
2DK	1	10.82.0.0/17	89.77.67.25/30
	2	10.82.128.0/17	89.77.67.30/30
	3	10.83.0.0/17	89.77.67.33/30
	4	10.83.128.0/17	89.77.67.38/30
	5	10.84.0.0/17	89.77.67.41/30
	6	10.84.128.0/17	89.77.67.46/30
	7	10.85.0.0/17	89.77.67.49/30
2DL - E	1	10.85.128.0/17	89.77.67.54/30
	2	10.86.0.0/17	89.77.67.57/30
	3	10.86.128.0/17	89.77.67.62/30
	4	10.87.0.0/17	89.77.67.65/30
	5	10.87.128.0/17	89.77.67.70/30
	6	10.88.0.0/17	89.77.67.73/30
	7	10.88.128.0/17	89.77.67.78/30
2NA	1	10.89.0.0/17	89.77.67.81/30
	2	10.89.128.0/17	89.77.67.86/30
	3	10.90.0.0/17	89.77.67.89/30
	4	10.90.128.0/17	89.77.67.94/30
	5	10.91.0.0/17	89.77.67.97/30
	6	10.91.128.0/17	89.77.67.102/30
	7	10.92.0.0/17	89.77.67.105/30
2NB	1	10.92.128.0/17	89.77.67.110/30
	2	10.93.0.0/17	89.77.67.113/30
	3	10.93.128.0/17	89.77.67.118/30
	4	10.94.0.0/17	89.77.67.121/30
	5	10.94.128.0/17	89.77.67.126/30
	6	10.95.0.0/17	89.77.67.129/30
	7	10.95.128.0/17	89.77.67.134/30